Generative AI

College of Computing

Final Projects topics

29 January 2026

Final project structure

- Complete a project in teams of 2-3 students
- Projects are to be chosen from the topics' list in this document
- More than one group of 2-3 students can choose the same topic, provided the final solutions presented are different
- Proposal choice: 08 February 2026 at 11:59pm. There are Two classes of projects:
  - Train or fine-tune a medium-sized LLMs based solution for the application at hand: HuggingFace hub
  - Evaluate different LLMs and understand their capabilities, limitations and risks on the application chosen
- Proposed applications: Agriculture, Supply chain, Innovation and Management, Geosciences

LLMs based systems

Projects - Motivation

- Challenges for LLMs-based systems in real-world applications such as Agriculture, Geosciences, Supply Chain and more are:
  - Data availability and quality
  - High computational costs
  - Infrastructure and technological barriers
  - Domain-Specific complexity
  - Ethical and social Concerns
  - Lack of transparency and explainability

Projects aims

- Solutions to data availability and quality
- Solutions to domain specific complexity
- Solutions to high computational costs

Projects - Motivation

- To address some of the challenges on data availability, domain specific complexity or high computational complexity, several LLMs based solutions have been proposed in both scientific research and commercial environments
- Examples of these solutions are:
  1. Extracting data from available scientific research articles or public resources in a specific domain to deal with the challenge of data quality and availability
  2. Developing language models leveraging domain specific data for a specific application purpose such as prediction, monitoring, improving detection and, etc...
  3. Developing SML (Small Language Models) ideas for specific applications

Projects - Motivation

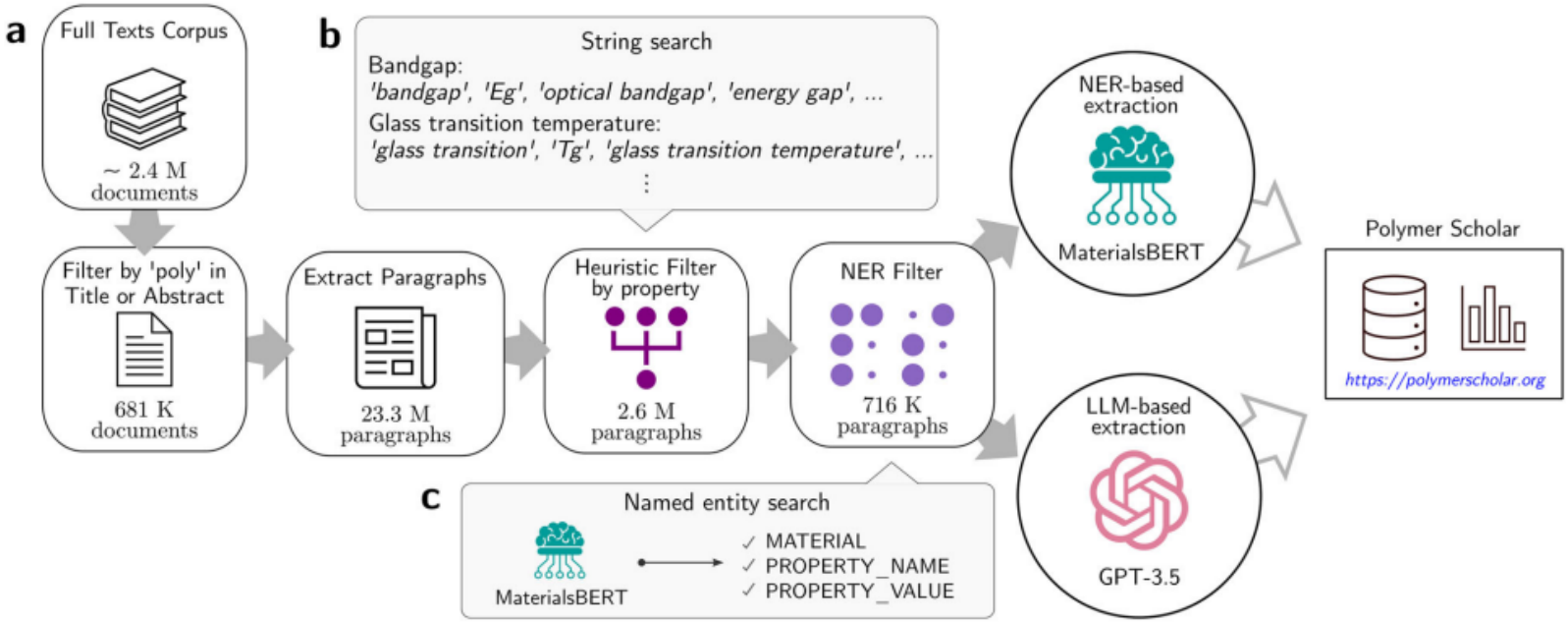
- Solutions examples for real-world application: Agriculture

Application Area	Research Study	Key Contributions	Type of Model
Precision Farming	Peng et al. (2023)	Explored the use of embedding-based retrieval models integrated with LLMs to optimize irrigation, fertilization, and crop disease monitoring through real-time, data-driven insights.	Embedding-based retrieval models
	Silva et al. (2023)	Demonstrated how GPT-4 can provide agronomist-level advice, assisting farmers in precision decision-making through better resource management.	GPT-4
Crop Disease Detection	Rezayi et al. (2022)	Developed AgriBERT, a language model leveraging domain-specific agricultural data for plant health monitoring, improving disease detection and prevention.	AgriBERT (BERT-based model)
	Yang et al. (2024)	Introduced PLLaMa, an LLM tailored for plant science, enhancing crop disease detection through text and image-based symptom analysis.	PLLaMa (Specialized LLM)
	Wang et al. (2023)	Developed an intelligent question-answering system for real-time crop disease diagnostics and pest management in fruits and vegetables.	Question-answering model (LLM)
Weather Forecasting	Peng et al. (2023)	Demonstrated how LLMs can analyze historical weather data, enhancing the accuracy of short-term weather forecasts and improving irrigation and planting schedules.	Embedding-based retrieval models
	Wang et al. (2023)	Created a question-answering system to provide weather-related recommendations and help farmers adapt to climate change impacts on agriculture.	LLM for QA (GPT-3 based)
Farmer Advisory Systems	Zhao et al. (2023)	Showed how ChatGPT could classify agricultural texts and provide multilingual advisory services for farmers, breaking down language barriers.	ChatGPT (GPT-3.5)
	Silva et al. (2023)	Demonstrated how GPT-4 can assist with complex agronomic queries, providing personalized advice to farmers based on local conditions and challenges.	GPT-4
Supply Chain Optimization	Peng et al. (2023)	LLMs were applied to optimize agricultural supply chains by integrating data from logistics, markets, and weather conditions to reduce wastage and improve transport efficiency.	Embedding-based retrieval models
	Wang et al. (2023)	Applied LLMs in demand forecasting, price prediction, and inventory management, optimizing farm-to-market logistics and ensuring timely delivery of crops.	GPT-based model for forecasting
Policy Recommendations	Peng et al. (2023)	Used LLMs to synthesize complex agricultural reports and recommend data-driven policies on climate adaptation, food security, and resource management.	Embedding-based retrieval models
	Yang et al. (2024)	Explored scenario modeling for governments using LLMs to predict outcomes of various agricultural policies, aiding in risk assessment and long-term planning.	Specialized LLM for scenario modeling

Jetir

Projects - Motivation

- Solutions examples for real-world application: Material Sciences
  - **Data extraction from polymer literature using LLMs**



Material

Projects - Direction

- Students who choose this theme can consider the following directions for their projects:
  - Developing a similar LLM-based system for other application area (s): Geosciences, Supply-chain, Management and Social Sciences, Agriculture
  - Investigating a different type of LLM for a previously proposed solution
  - Developing a **chatbot-like** system from different available scientific research articles to tackle a specific application's need (different than what is proposed in the litterature)
  - Proposing their own idea using LLMs-based systems to tackle one of the common challenges to many real-world applications:
    - Data availability and quality
    - Domain specific complexity
    - High computational costs

GraphRAG domain specific LLMs

Projects - Motivation

- LLMs have shown remarkable capabilities in various tasks: text comprehension, question answering and content generation
  - Application to specialised domains remains challenging: need for deep expertise
- Retrieval-Augmented Generation (RAG): A promising solution to LLMs customisation for professional fields RAG
  - Difficulties in knowledge integration and system efficiency bottlenecks at scale
- Graph Retrieval-Augmented Generation (GraphRAG) has recently emerged as a new paradigm GraphRAG
  - Leverage structured knowledge: organise and retrieve information
  - A subclass of the RAG framework: leverages graph structures to organise and retrieve knowledge

Projects - Motivation

- Some open-source projects and frameworks for GraphRAG development and deployment:
  - Microsoft GraphRAG ▸ GraphRAG
  - Nano-GraphRAG ▸ NanoGraphRAG
  - Fast GraphRAG ▸ FastGraphRAG
  - LightRAG ▸ LightRAG
- Some initial applications in Medical field: ▸ MedicalGRAGpaper ▸ BiomedicalfbmRAG
- Some initial applications in Geosciences and Mining: ▸ CoalMineGraphRAG ▸ GeosciencesGraphRAG

Projects - Direction

- Students who choose this theme can consider the following directions for their projects:
  - Developing a similar GraphRAG LLM-based system for other application area (s): Geosciences, Supply-chain, Management and Social Sciences or Agriculture
  - Investigating a different type of GraphRAG LLM-based variant for a previously proposed solution/application
  - Developing a **chatbot-like** system from different available scientific research articles to tackle a specific application's need
  - Proposing their own idea using GraphRAG LLMs-based systems to tackle one of the common challenges to many real-world applications:
    - Data availability and quality
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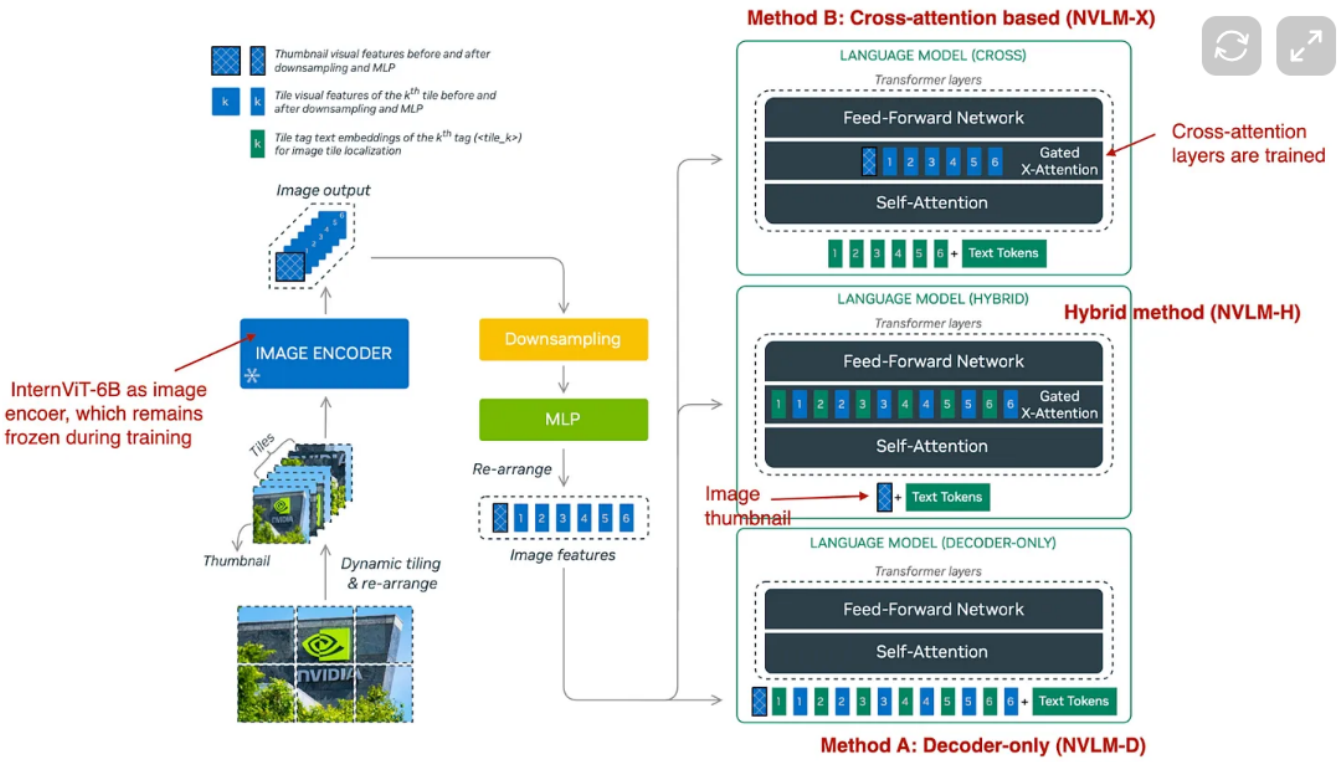
Multimodal LLMs

Projects - Motivation

- A multimodal LLM, or MLLM: process and reason across multiple types of data or modalities such as text, images and audio ▸ MultimodalLLMs
- Training multimodal LLMs involves:
  - Careful designing of data collection processes
  - Suitable choice of training strategies
  - Data quality to develop models: understanding and reasoning across multiple modalities
- Different scientific Multimodal LLMs and commercial solutions have been presented so far, making advantage of fusing different types of data and modalities

Projects - Motivation

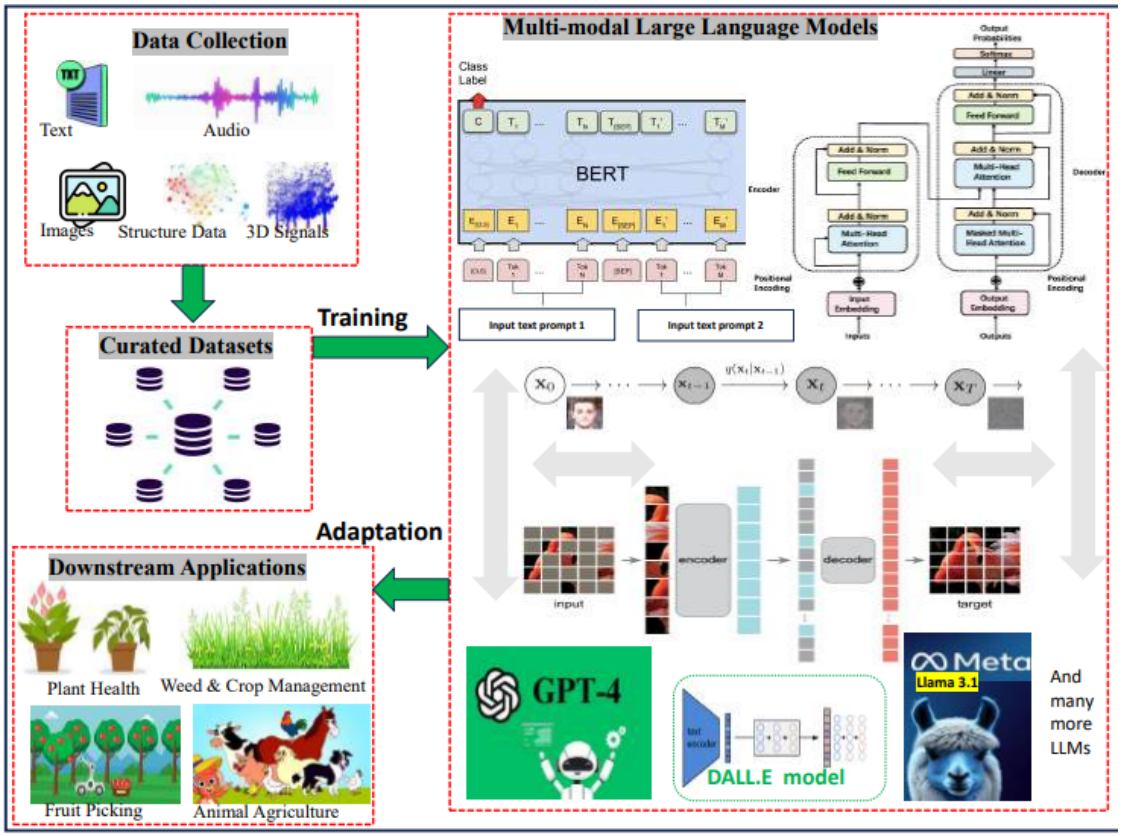
- Technical MLLMs development example:
  - **NVLM - Open Frontier-Class Multimodal LLMs:**



NVLM

Projects - Motivation

- MLLMs application example:
  - **Multimodal LLM in Agriculture:**



LLMAgriculture LLMAgriculture

Projects - Direction

- Students who choose this theme can consider the following directions for their projects:
  - Developing a similar Multimodal LLM-based system for other application area (s): Geosciences, Supply-chain, Management and Social Sciences or Agriculture
  - Investigating a different type of a Multimodal LLM-based variant for a previously proposed solution/application
  - Developing a **chatbot-like** system from different available scientific research articles to tackle a specific application's need
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Good Luck!